FAA FAARFIELD CM Report — Computational Mechanics

Print Report

FAARFIELD 2.1.1 (Build 03/17/2026)

Job: New Job 1**Structure:** New Structure 1**Analysis:** HMA on Aggregate**Generated:** 2026-03-17 22:29:14

MAX TOTAL CDF

0.996880

DESIGN THICKNESS

9.75 in.

AIRCRAFT IN MIX

1

CRITICAL OFFSET

190 in.

SUBGRADE MODULUS

15,000 psi

CONVERGED / ITERATIONS

Yes / 3

Table of Contents

Pavement Structure Summary

Design Layers

1	4.00	200,000	1
2	16.02	68,847	6
3	9.75	20,418	8
4	Semi-infinite	15,000	4

Expanded Sublayer Structure (after modulus adjustment)

1	4.00	200,000	1
2	6.02	75,474	6
3	10.00	64,858	6
4	9.75	20,418	8
5	Semi-infinite	15,000	4

Evaluation Depth at Subgrade = 29.77 in.

Unbound Aggregate Modulus — Sublayering Procedure

Unbound aggregate layers (crushed base and uncrushed subbase) do not have a single fixed modulus. FAARFIELD subdivides each aggregate layer into sublayers and computes a depth-dependent modulus for each using an empirical formula. The modulus of each sublayer depends on the modulus of the material below it — sublayers near the bottom (close to the subgrade) have lower moduli, while sublayers near the top have higher moduli. The computation proceeds from the bottom of the aggregate layer upward.

Sublayer Modulus Reduction Formula

$$f_1 = 1 + C \times \ln(t) / \ln(10)$$

$$f_2 = D \times \ln(E_{i-1}) \times \ln(t) / \ln^2(10)$$

$$E_i = E_{i-1} \times (f_1 - f_2)$$

where t = sublayer thickness (in.), E_{i-1} = modulus of the layer below (psi). Applied iteratively from the bottom sublayer upward.

C (thickness factor)	10.52	6.88
D (modulus interaction)	2.00	1.56
E_{below} (psi)	20,418	15,000
Number of sublayers	2	1

P-209 Crushed Aggregate Base — Sublayer Moduli

1 (top)	6.02	75,474
2 (bottom)	10.00	64,858
↓ *Layer below*	—	*20,418*

P-154 Uncrushed Aggregate Subbase — Sublayer Moduli

1 (top)	9.75	20,418
↓ *Layer below*	—	*15,000*

Modulus vs. depth profile for the expanded sublayer structure. Aggregate layers are subdivided and their moduli computed bottom→up using the empirical reduction formula. The teal step line traces the modulus at each sublayer depth.

Design Equations

Subgrade Strain Failure Model (FAA Standard)

$$AA = 0.000247 + 0.000245 \times \log_{10}(E_{subgrade})$$

$$BB = 0.0658 \times E_{subgrade}^{0.559}$$

$$N_{fail} = 10{,}000 \times (AA / \varepsilon_v)^{BB}$$

For this structure ($E_{subgrade}$ = 15,000 psi): AA = 0.001270, BB = 14.212

Cumulative Damage Factor (CDF)

$$CDF_{aircraft} = \text{Repetitions} \times (C/P) / N_{fail}$$

$$CDF_{total} = \Sigma\, CDF_{aircraft} \quad \text{(summed over all aircraft)}$$

where C/P = Coverage-to-Pass ratio from Gaussian lateral wander model (σ = 30.435 in.)

Coverage-to-Pass (Gaussian Wander Model)

$$C/P(\text{offset}) = \int G(x; \sigma)\, dx \quad \text{over tire contact width}$$

$$G(x; \sigma) = [1 / (\sigma\sqrt{(2\pi)})] \times \exp(-x^2 / 2\sigma^2)$$

σ = 30.435 in. (std. dev. of lateral wander)
Projected tire width at depth d: TW_{proj} = TW + 2d

Convergence Criterion

$$|\ln(CDF_{total})| < 0.005 \quad \text{(exit criterion)}$$

Design converges when $0.995 < CDF_{total} < 1.005$
Sublayers activated when 0.5 < CDF < 2.0 (|ln(CDF)| < 0.690)

Understanding Coverage-to-Pass (C/P)

The Coverage-to-Pass (C/P) ratio is the probability that a given evaluation strip (10 in. wide) on the pavement surface will be loaded by a passing wheel, accounting for Gaussian lateral wander of the aircraft about the nominal wheel path centerline. The standard deviation of wander is σ = 30.435 in. (corresponding to a wander width of 70 in.).
For multi-wheel gear configurations, the total C/P at each strip is the superposition (sum) of the individual GaussArea contributions from every wheel in the gear assembly, because the entire gear wanders as a rigid body. FAARFIELD evaluates 41 strips on one side of the nominal wheel path centerline (offsets 0 to 400 in.).

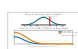

Figure 1: Gaussian lateral wander and single vs. dual wheel C/P comparison

Fatigue Characterization

The following chart shows the subgrade fatigue model curve (allowable repetitions vs. vertical strain) with each aircraft's computed strain and N_{fail} plotted as scatter points.

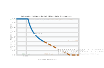

Figure 2: Subgrade fatigue model with aircraft scatter points

Aircraft Fatigue Parameters

B787-9	1328.79	0.004000	8.100	3.469E+04	24,000	1.45E+00	Bleasdale

Figure 3: Fatigue life reserve ratio per aircraft

Per-Aircraft Detailed Breakdown

Aircraft 1: B787-9

Gear Type	X	Landing gear configuration
Gear Load	563,000 lbs	Maximum gear load applied to pavement
Tire Pressure	229.0 psi	Inflation pressure
Tire Width (TW)	24.39 in.	Contact width at surface
Contact Area	614.63 in.^{2}	Static tire contact area
Annual Departures	1,200	Annual departure level (ADL)
Total Repetitions	24,000	Computed from ADL × 20-year design life
# Gear Loads	1	Main + belly gear load groups
Projected TW at Subgrade	83.92 in.	TW + 2×depth (45° spread)
Max Vertical Strain	1328.79 με	LEAF-computed subgrade strain
N_{fail}	3.469E+04	Allowable repetitions (Bleasdale)
Max C/P Ratio	0.72039	Peak coverage-to-pass ratio
Max CDF (this aircraft)	0.996880	Peak damage contribution
CDF at Critical Offset	0.996880	Damage at critical strip

Figure 4: Gear configuration for B787-9

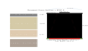

Figure 5: Pavement cross-section for B787-9

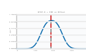

Figure 6: CDF by offset for B787-9

▶ CDF by Offset — B787-9 (click to expand)

Step-by-Step Computation

1. Apply gear load of 563,000 lbs with tire pressure 229.0 psi and tire contact width 24.39 in. (X gear).
2. Run LEAF to compute vertical subgrade strain. Result: ε_v = 1328.79 με at evaluation depth 29.77 in..
3. Compute N_{fail} using Bleasdale model: AA=0.004000, BB=8.100. N_{fail} = 3.469E+04.
4. Compute C/P at 41 strips. Peak C/P = 0.72039.
5. Project tire width to subgrade (45° spread): TW_{proj} = 83.92 in..
6. Total repetitions: 1,200 × 20 = 24,000.
7. CDF = Reps × C/P / N_{fail}. Max CDF = 0.996880. CDF at critical strip = 0.996880.

Coverage-to-Pass (C/P) Distribution

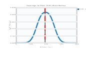

Figure 7: C/P ratio distribution across pavement width

CDF Sweep Table (41 offsets)

0	0.00000	0.000005	**0.000005**
10	0.00001	0.000017	**0.000017**
20	0.00005	0.000062	**0.000062**
30	0.00016	0.000216	**0.000216**
40	0.00050	0.000694	**0.000694**
50	0.00150	0.002076	**0.002076**
60	0.00410	0.005669	**0.005669**
70	0.01022	0.014144	**0.014144**
80	0.02319	0.032096	**0.032096**
90	0.04782	0.066178	**0.066178**
100	0.08965	0.124064	**0.124064**
110	0.15309	0.211850	**0.211850**
120	0.23873	0.330363	**0.330363**
130	0.34122	0.472180	**0.472180**
140	0.44922	0.621642	**0.621642**
150	0.54857	0.759123	**0.759123**
160	0.62734	0.868117	**0.868117**
170	0.68021	0.941286	**0.941286**
180	0.70927	0.981494	**0.981494**
190	0.72039	0.996880	**0.996880**
200	0.71773	0.993200	**0.993200**
210	0.70015	0.968874	**0.968874**
220	0.66218	0.916329	**0.916329**
230	0.59882	0.828655	**0.828655**
240	0.51075	0.706781	**0.706781**
250	0.40626	0.562184	**0.562184**
260	0.29882	0.413505	**0.413505**
270	0.20201	0.279542	**0.279542**
280	0.12497	0.172933	**0.172933**
290	0.07052	0.097584	**0.097584**
300	0.03622	0.050118	**0.050118**
310	0.01690	0.023393	**0.023393**
320	0.00717	0.009923	**0.009923**
330	0.00277	0.003832	**0.003832**
340	0.00098	0.001351	**0.001351**
350	0.00032	0.000436	**0.000436**
360	0.00010	0.000132	**0.000132**
370	0.00003	0.000037	**0.000037**
380	0.00001	0.000010	**0.000010**
390	0.00000	0.000003	**0.000003**
400	0.00000	0.000000	**0.000000**

Critical offset at #20 = 190 in., Max CDF = 0.996880

CDF Distribution Across Pavement Width

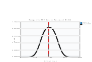

Figure 8: Composite CDF distribution across pavement width

Figure 9: CDF contribution per aircraft at critical offset

CDF Contribution Summary at Critical Offset

B787-9	0.996880	100.0%
Total	**0.996880**	**100.0%**

Newton-Raphson Convergence

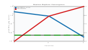

Figure 10: Newton-Raphson convergence history

Iteration Log

1	6.00	11.613960	2.45221	0.00	0.000	Yes
2	8.71	2.172299	0.77579	0.00	0.000	Yes
3	9.75	0.996880	0.00312	0.00	0.000	Yes

Convergence Summary